March 25, 2025

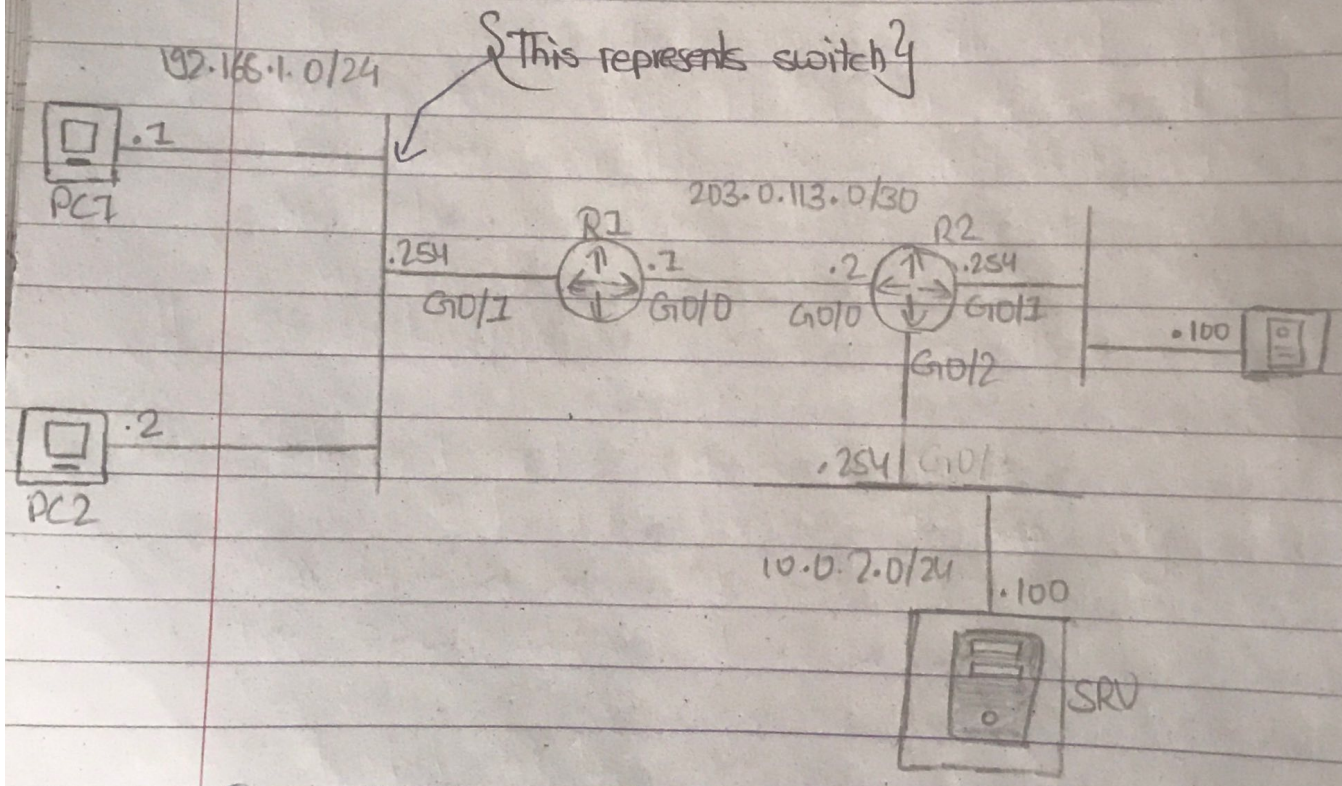
Day-34

Standard Access Control Lists

- ACLs (Access Control Lists) have multiple uses.
- In Day 34 and Day 35, we will focus on ACL from security purpose.

- ACLs function as a packet filter, instructing the router to permit or discard specific traffic.
- ACLs can fu filter traffic based on src/dst IP addresses src/dst Layer 4 ports etc. {We'll focus on IP & port for CCNA}

- ~~ACLs can filter traffic based on source~~



We shouldn't just configure ACLs randomly, we must use them to achieve a certain requirement

For eg. network policy in is: (i) Hosts in 192.168.1.0/24 can access the 10.0.1.0/24 network (ii) Hosts in 168.2.0/24 can't access the 10.0.1.0/24 network.

NOB:

ACLs are configured globally on the router. (global config mode)
They are an ordered sequences of ACEs (Acces Control Entries)

ACL 1.
1) If src IP = 192.168.1.0/24 then permit
2) If src IP = 192.168.2.0/24 then deny
3) If src IP = any, then permit.

- Configuring an ACL in global config mode won't make the ACL take effect
- The ACL must be applied to an interface
- ACLs are applied either inbound or outbound
Inbound - Incoming traffic Outbound - Outgoing traffic.

- ACLs are made up of one or more ACEs.
- When the router the checks a packet against the ACL, it processes the ACEs in order, from top to bottom. If the top gets self satisfied it won't check that of below.

- If the packet matches @ one of the ACEs in the ACL, the router takes the action and stops processing the ACL. All entries below the matching entry will be ignored.

Scanned with CamScanner

NOB: A maximum of one ACL can be applied to a single interface per direction.
Inbound: Max one ACL
Outbound: Max one ACL

Implicit Deny

What happens if a packet doesn't match any of the entries in an ACL?

↳ The router will deny the packet, it won't forward it.

- There's an "implicit deny" at the end of all ACLs
- The 'implicit deny' tells router to deny all traffic that doesn't match any of the configured entries in ACL

ACL Types

① Standard ACLs.
Match based on src IP add. only
a) Standard numbered ACLs
b) Standard Named ACLs.

② Extended ACLs
Extended numbered ACLs match based on src/dst IP, src/dst port etc.
a) Extended numbered ACLs
b) Extended named ACLs

Standard ACLs

Standard Numbered ACLs

- Standard ACLs match traffic based only on the src IP add. of the packet. {The router doesn't check dst IP, dst port etc.}

- Numbered ACLs are identified with a number (i.e ACL1, ACL2 etc)
- Different types of ACLs have a different range of numbers that can be used.
  - ↳ Standard ACLs can use 1-99 and 1300-1999

{IP ACL is the type of ACL we've to learn for the CCNA}

- The basic command to configure a standard numbered ACL is:

R1(config)# access-list number {deny|permit} ip wildcard-mask.

{1-99 or 1300-1999}

Eg.

R1(config)# access-list 1 deny 1.1.1.1 0.0.0.0 (It denies single host)

{Instead we can just write access-list 1 deny 1.1.1.1 & router'll understand. or we can use keyword "**host**" before IP add.

Remark

{It's like 'interface description'}

R1(config)# access-list 1 remark ## BLOCK BOB FROM SERVER ##

- {It doesn't have any effect on ACL, its just a description that helps us remember the purpose of ACL when looking at in configuration.
  # has nothing to do.
  The command is

R1(config)# access-list number remark remark_we_write

Other commands & stuffs

- R1(config)# do show access-lists
  Standard IP access list 1
  10 deny 1.1.1.1
  20 permit any
  ↳ The order of entry. It's very important. If permit any was at first, all traffic would've been permitted and the deny entry would be useless. (Modern devices mostly prevent us from doing these)

- R1(config)# do show ip access-lists
  Earlier command would've shown all access-list but this command would only show ip access-list

- R1(config)# do show running-config | include access-list

Applying ACL to interface.

R1(config)# int g0/0
R1(config-if)# ip access-group number (inbound)

In our network diagram, suppose the condition is
- PC1 can access 192.168.2.0/24
- Others PCs in 192.168.1.0/24 can't access 192.168.2.0/24

- R1(config-if)# ip access-group 1 out.

NOB: Standard ACLs should be applied as close to the destination as possible.

Standard Named ACLs.

- Standard ACLs match traffic based only on the source IP address of the packet.
- Named ACLs are identified with a name (ie 'BLOCK_BOB')
- Standard named ACLs are configured by entering 'standard named ACL config mode', and then configuring each entry within that config mode.

R1(config)# ip access-list standard <acl-name>
R1(config-std-nacl)# [entry-number] {deny|permit} ip <source> <wildcard-mask>

- Code gives no compulsion to mention entry-number. If not mentioned, it goes like 10, 20, 30 -- etc.

Some show commands

- R1(config)# int g0/0
- R1(config-if)# ip access-group name-of-ACL in/out
- show access-lists
- show running-config | section access-list

Some Important Deep but not necessary for CCNA

- The router may reorder the 132 entries
- This improves the efficiency of processing the ACL.
- It doesn't change effect of ACL.
- It applies to both named & numbered standard ACLs
- Packet tracer does not do this